

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (CAD/CAM) Examination

May 2013

ME 709.01 Advance Engineering Optimisation Techniques

Date: 15.05.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

- Q - 1 (a)** Five strip-layout selection attributes viz. economical material utilization (Ur), die cost (Dc), [09]  
stamping operational cost (Oc), required production rate (Pr) and job accuracy (Ja) are  
presented in Table 1 and the estimated quantitative values are assigned to Ur, Dc, Oc, Pr and  
Ja. Find the optimum layout using Modified TOPSIS method.

Table 1: Strip layout attributes

| Layout | Ur   | Dc    | Oc  | Pr  | Ja |
|--------|------|-------|-----|-----|----|
| (a)    | 0.26 | 25000 | 130 | 80  | 4  |
| (b)    | 0.40 | 28560 | 138 | 120 | 3  |
| (c)    | 0.33 | 31109 | 90  | 150 | 3  |
| (d)    | 0.32 | 31702 | 150 | 125 | 2  |
| (e)    | 0.31 | 32390 | 160 | 110 | 2  |
| (f)    | 0.31 | 32663 | 116 | 108 | 2  |

Ur is expressed in %; Dc in Rupees; Oc in Rupees/1000 pieces; Pr in pieces/minute.

The normalized weights of each attribute are:  $W_{Ur} = 0.2815$ ;  $W_{Dc} = 0.1054$ ;  $W_{Oc} = 0.0501$ ;  
 $W_{Pr} = 0.2815$ ; and  $W_{Ja} = 0.2815$ .

- Q - 2 (a)** State Kuhn Tucker's conditions. For the following problem check whether constraint qualification and Kuhn Tucker conditions are satisfied or not at point (0,0). [07]

$$\text{Minimise } f(x_1, x_2) = (x_1 + 1)^2 - x_2$$

subjected to

$$g_1(x_1, x_2) = x_1^3 - 2x_2 \leq 0$$

$$g_2(x_1, x_2) = x_1^3 + 2x_2 \leq 0$$

- (b)** Determine whether the following function is concave, convex or neither [07]

$$f(x) = x_1^2 + 3x_1x_2 + 2x_2^2$$

**OR**

- Q - 2 (a)** Solve the following function using Lagrange's Multiplier [07]

$$\text{Maximise } Z = 5x_1 + x_2 - (x_1 - x_2)^2$$

$$\text{Subject to } x_1 + x_2 = 4,$$

$$x_1, x_2 \geq 0$$

- (b)** Determine whether the following function is concave, convex or neither [07]

$$f(x) = x_1x_2$$

- Q - 3 (a)** Define fuzzy sets and membership function. Explain different membership functions available for MF formulation. [04]

- (b)** Explain the terms 'Cross-over' and 'Mutation' in Genetic Algorithm with examples. [04]

- (c)** Describe the architecture of a typical Artificial Neural Network. [04]

**OR**

- Q - 3 (a)** Describe four steps in building a fuzzy system. [04]

- (b)** What are the components of Genetic Algorithm? Describe them in brief. [04]

- (c)** Give a brief comparison of Biological and Artificial Neural Network. [04]

## SECTION – II

**Q - 4 (a)** Define Local and Global Optimum. Also discuss about inflexion point. [04]

**(b)** Find maxima and minima, if any, of the function [04]

$$f(x) = 4x^3 - 18x^2 + 27x - 7$$

**Q - 5 (a)** Compare different elimination methods based on final interval of uncertainty. [04]

**(b)** The objective function is: [08]

$$F(x) = 2(x^{0.5} - x) + 0.01x^{0.5} \quad x \in [0.15, 0.31]$$

Find the maxima of above function with precision of  $\pm 0.02$  with golden section method.

## OR

**Q - 5 (a)** Define unimodal function? Classify methods used to approximate solution of such a function. [04]

**(b)** Solve the following using Fibonacci Method in the interval (0,3) for  $n=8$  [08]

$$f(x) = \frac{0.5}{\sqrt{1+\lambda^2}} - \sqrt{1+\lambda^2} \left(1 - \frac{0.5}{\sqrt{1+\lambda^2}}\right) + \lambda$$

**Q - 6** Attempt the following (Any three) [15]

**(a)** Algorithm of Powell's Method

**(b)** C Program on Newton Raphson's Method

**(c)** Flowchart of Fibonacci Method

**(d)** Algorithm on Steepest Decent Method

**(e)** C Program on Exhaustive Search Method

**(f)** Algorithm of Hooke and Jeeves' Method

\*\*\*\*\*